

PROJECT DESIGN PRESENTATION

FutureVerse: A Machine Learning-Based Framework for Sugarcane Growth Prediction and Intelligent Farming Recommendation

GROUP NO:	**A18**

GUIDE : Ms. V. V. Manglaram

FutureVerse

A Machine Learning-Based Framework for Sugarcane Growth Prediction and Intelligent Farming Recommendations

Problem Statement

- Sugarcane growth and yield depend on soil pH, N-P-K, temperature, humidity, rainfall, irrigation and crop age.
- Field conditions vary, making it difficult to combine all factors for reliable prediction.
- Farmers need simple, data-driven guidance rather than isolated technical analysis.

Proposed Solution

- Use machine learning to predict sugarcane growth (%) and estimated yield (tons/hectare).
- Compare multiple regression models and select the best-performing model using validation metrics.
- Convert predictions into fertilizer, irrigation, crop-management and soil-improvement recommendations.

Team Members

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Project Guide

- Ms. V.V. Manglaram
- Assistant Professor

Expected Outcome

- Farmer-friendly web decision-support system
- Prediction + practical recommendations

2. OVERALL SYSTEM ARCHITECTURE

Dataset
- Historical agricultural data
- Soil • Weather • Crop

Preprocessing
- Validate inputs
- Handle missing values
- Normalize / select features

Model Training
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

Evaluation
- MAE
- RMSE
- R^2
- Select best model

Saved Model
- Best model + preprocessor
- Ready for prediction

RUNTIME / FARMER WORKFLOW

1. Farmer Input
- Soil pH & NPK
- Weather
- Irrigation
- Crop age

2. REST API
- FastAPI backend
- Validate request
- Return results

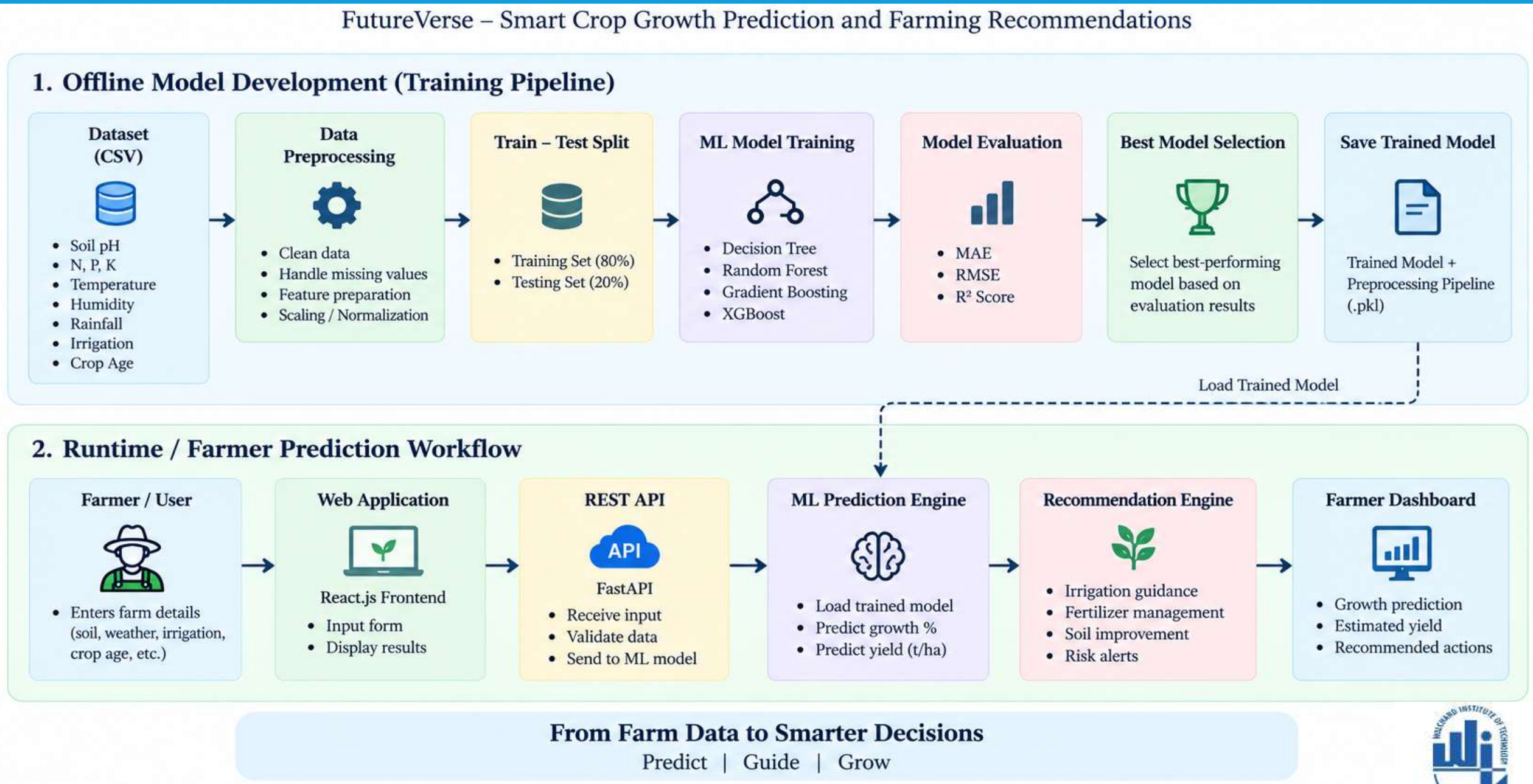
3. Preprocessing
- Same trained pipeline
- Transform features

4. Prediction
- Selected ML model
- Growth (%)
- Yield (t/ha)

5. Recommendations
- Fertilizer
- Irrigation
- Crop / soil advice

Farmer Dashboard
- Growth % | Estimated Yield (t/ha) | Farm Status | Recommended Actions

2. OVERALL SYSTEM ARCHITECTURE



3. MODULE DESIGN

1 User Input Module

Collect soil pH, N, P, K, temperature, humidity, rainfall, irrigation and crop age.

2 Validation & Preprocessing

Validate values, handle missing data, normalize numerical features and select relevant inputs.

3 ML Prediction Engine

Train/use Decision Tree, Random Forest, Gradient Boosting and XGBoost regression models.

4 Best Model Selection

Evaluate MAE, RMSE and R^2; select the best-performing model on validation data.

5 Recommendation Engine

Generate fertilizer, irrigation, crop-management and soil-improvement guidance.

6 Result Generation

Display predicted growth, estimated yield and recommendations through the dashboard.

4. DATA / DATABASE / WORKFLOW DESIGN

Dataset Structure

• Input features:
• Soil pH • Nitrogen • Phosphorus • Potassium
• Temperature • Humidity • Rainfall
• Irrigation • Crop age
•
• Target outputs:
• Sugarcane Growth (%)
• Estimated Yield (tons/hectare)
•
• Data handling:
• Validation • Missing values
• Normalization / scaling • Feature selection

DATA-PROCESSING PIPELINE

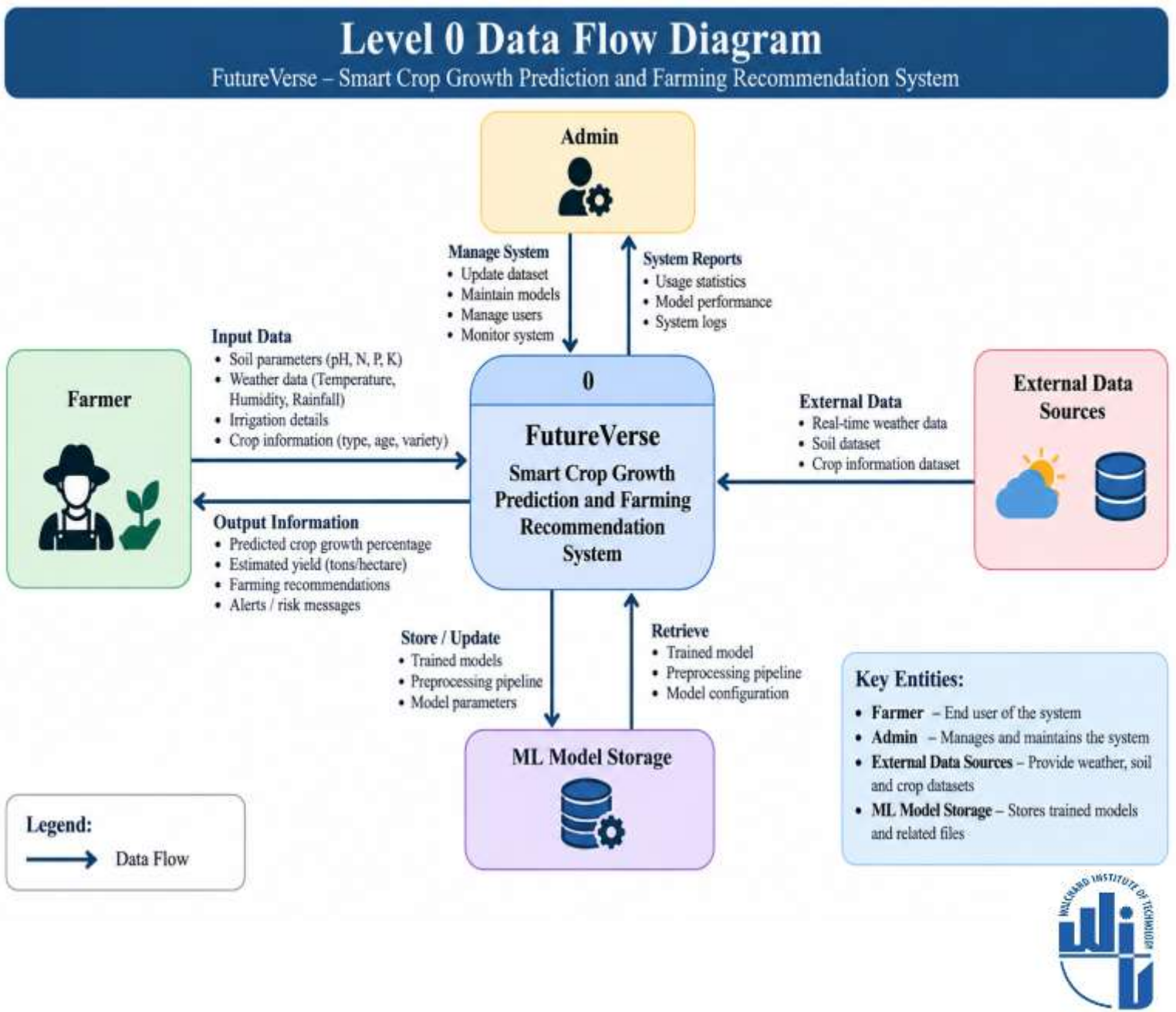
1	**Load Dataset**	Historical agricultural records
2	**Clean & Validate**	Check ranges and missing values
3	**Prepare Features**	Normalize / select relevant inputs
4	**Split Data**	Training and validation/test sets
5	**Train Models**	Four regression algorithms
6	**Evaluate & Select**	MAE, RMSE, R^2
7	**Save Pipeline**	Selected model + preprocessing
8	**Predict & Recommend**	Growth, yield and actions

Core prediction workflow does not require a traditional database for permanent farmer/crop storage; inputs can be processed at runtime.

4. DATA / DATABASE / WORKFLOW DESIGN

DATA-PROCESSING PIPELINE

1	**Load Dataset**	Historical agricultural records
2	**Clean & Validate**	Check ranges and missing values
3	**Prepare Features**	Normalize / select relevant inputs
4	**Split Data**	Training and validation/test sets
5	**Train Models**	Four regression algorithms
6	**Evaluate & Select**	MAE, RMSE, R^2
7	**Save Pipeline**	Selected model + preprocessing
8	**Predict & Recommend**	Growth, yield and actions



5. CORE TECHNICAL DESIGN

Machine Learning Prediction

1. Prepare input features from soil, weather, irrigation and crop age.
2. Train four regression models:
• Decision Tree Regressor
• Random Forest Regressor
• Gradient Boosting Regressor
• XGBoost Regressor
3. Evaluate using MAE, RMSE and R^2.
4. Select the best-performing model empirically.
5. Predict Growth (%) and Estimated Yield (t/ha).

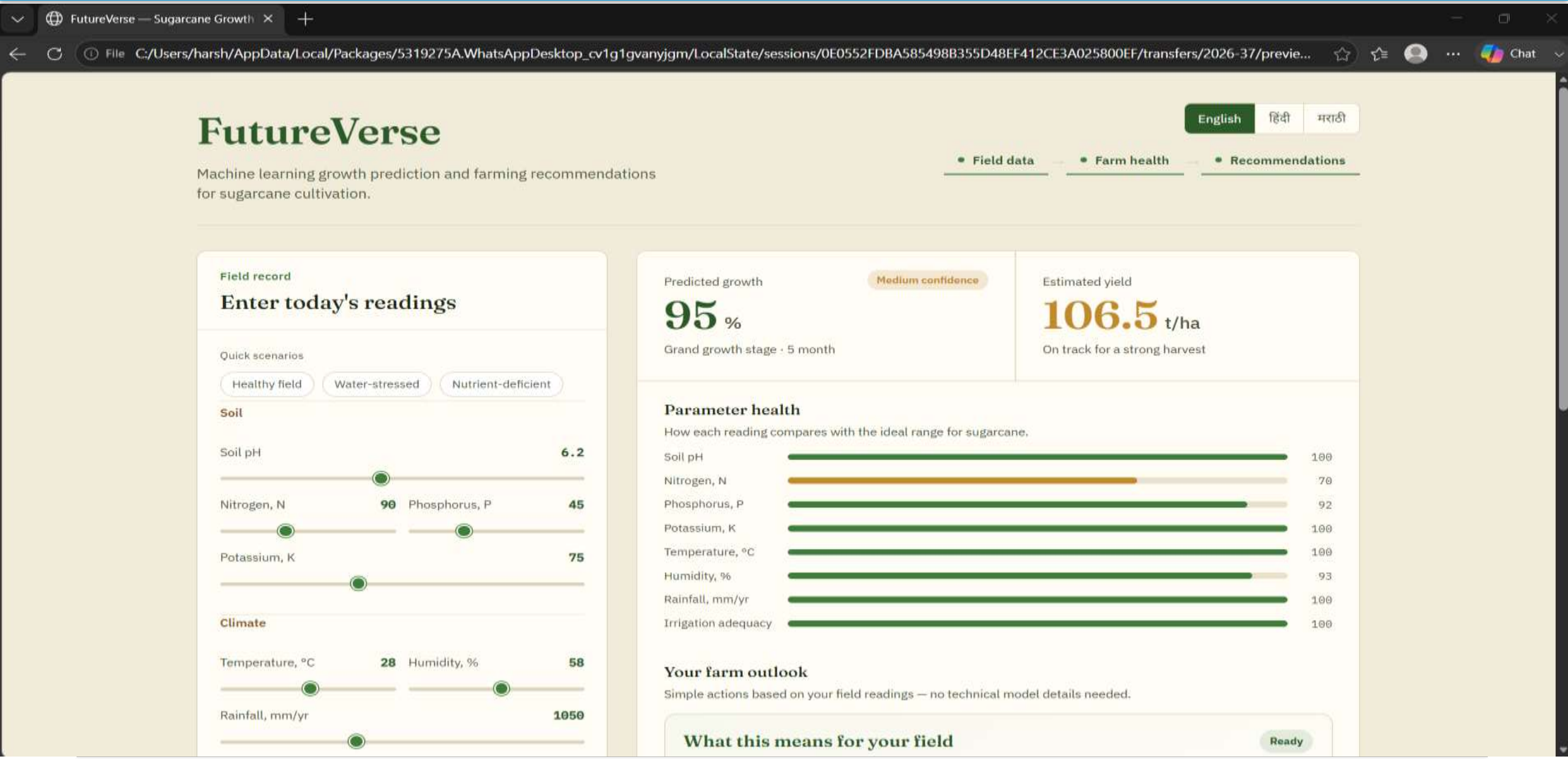
Intelligent Recommendation Engine

Prediction results + field conditions
• → Fertilizer application guidance
• → Irrigation scheduling guidance
• → Crop management practices
• → Soil improvement strategies

REST API & Runtime Flow

• React frontend sends farmer inputs to the REST API.
• Backend validates and preprocesses the request using the trained pipeline.
• Selected ML model generates prediction.
• Recommendation engine converts results into understandable actions.
• Dashboard displays the final output.

6. UI / PROTOTYPE / WIREFRAMES



FutureVerse — Sugarcane Growth
File C:/Users/harsh/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/0E0552FDBA585498B355D48EF412CE3A025800EF/transfers/2026-37/previe...
Chat

Rainfall, mm/yr **1050**

Irrigation & crop

Irrigation adequacy

Adequate

Soil type

Clay

Crop age, months **5**

Analyze growth

Simple actions based on your field readings — no technical model details needed.

What this means for your field

Ready

Focus on the areas that can improve your crop first.

FIELD STATUS	TOP PRIORITY	NEXT STEP
–	–	–

Recommendations

Generated from the gap between current readings and target ranges for sugarcane.

Fertilizer

Apply a balanced NPK dose to raise nitrogen — currently 90 kg/ha, target 120–180; and add phosphorus — currently 45 kg/ha, target 50–80.

Irrigation

clay soil retains water — space out irrigation to avoid waterlogging.

Crop management

the crop is in its grand growth stage; this is the main period for nutrient uptake — keep fertilizer and water on schedule.

Soil improvement

soil pH (6.2) is within the ideal range for sugarcane. Adding organic compost or farmyard manure will improve long-term soil structure and nutrient retention.

Download report

Readings are processed for this session only and are not stored. Figures are illustrative estimates for demonstration purposes, not agronomic guarantees.

7. TECHNOLOGY STACK & IMPLEMENTATION STATUS

Frontend

- React.js
- JavaScript ES6+
- HTML5
- CSS3
- Bootstrap
- Interactive dashboard

Backend & ML

- Python 3.11
- FastAPI / RESTful API
- Scikit-learn
- XGBoost
- Pandas & NumPy
- Matplotlib
- Joblib

Data / Platform

- CSV / dataset files for training
- Runtime processing of farmer inputs
- No traditional database required for core workflow
- Windows 11 / Ubuntu 22.04
- Client-server architecture

QUESTIONS

THANK YOU !